

Detection and Segmentation using Detectron2

Business Objective

Antibiograms are often used by clinicians to assess local susceptibility rates to select empiric antibiotic therapy, and monitor resistance trends over time. The testing occurs in a medical laboratory and uses culture methods that expose bacteria to antibiotics to test if bacteria have genes that confer resistance. Culture methods often involve measuring the diameter of areas without bacterial growth, called zones of inhibition. The zone of inhibition is a circular area around the antibiotic spot in which the bacteria colonies do not grow. The zone of inhibition can be used to measure the susceptibility of the bacteria to the antibiotic. The process of locating the zones and related inhibitions can be automated using image detection and segmentation.

Image Classification helps us classify what is contained in an image, whereas object Detection specifies the location of multiple objects in the image. On the other hand, image Segmentation will create a pixel-wise mask of each object in the images, which helps identify the shapes of different objects in the image.

We have dealt with image classification in the project, [Build a Multi Class Image Classification Model Python using CNN](#). This project will perform image detection and segmentation on a given set of images to detect the zones and inhibition of the bacteria present in a collection of images using the Detectron2 model.

Data Description

The dataset contains several antibiogram test images divided into training and testing folders. There are around 50 images in the training dataset and five in the testing.

Aim

To build a Detectron2 model that can detect zones and inhibitions in a given image.

Tech stack

- Language - Python
- Libraries - torch, detectron2, cv2, matplotlib, numpy

Approach

1. Import Detectron2 using the git command
2. Importing the required libraries.
3. Setup the input, output, train, test and annotations.json path
4. Register the coco instances with register name and annotations and form a dataset with configurations (in metadata catalog)
5. Data Visualization
6. Training the Detectron2 model
 - Define a class for coco Trainer
 - Get the default configurations from cfg and add the necessary parameters
 - Load the pre-trained model weights.
 - Perform training on the data.
7. Inference model
 - Set the default config file and add the required parameters
 - Define an inference image
 - Load the weights of trained model
 - Define default predictor object
 - Make the predictions and visualize the results.
 - Interpret the generated results
 - Plot the bounding boxes and masks in a given image

Modular code overview

```
input
|_Data
|   |_annotations
|       |_annotations.json
|   |_images
|       |_train
|       |_test

src
|_Engine.py
|_ML_Pipeline
|   |_admin.py
|   |_inference.py
|   |_training.py

lib
|_detection_segmentaion.ipynb

output
|_model_final.pth
```

Once you unzip the modular_code.zip file, you can find the following folders within it.

1. Input
2. Source Folder
3. Output
4. Lib

1. Input folder - It contains all the data that we have for analysis. We have two folders. Training_data and testing_data. The following three subfolders are present

- driving_license
- social_security
- others

2. Source folder - This is the most important folder of the project. This folder contains all the modularized code for all the above steps in a modularized manner. This folder consists of:

- Engine.py
- ML_Pipeline

The ML_Pipeline is a folder that contains all the functions put into different python files, which are appropriately named. These python functions are then called inside the engine.py file.

3. Output folder - The output folder contains the fitted model that we trained for this data. This model can be easily loaded and used for future use, and the user need not have to train all the models from the beginning.

4. Lib - This folder contains two notebooks.

- training_classification.ipynb
- Inference.ipynb

There is a ppt present in the reference subfolder. This is the one used during the presentation.

Project Takeaways

1. What is an antibiogram test?
2. What are zones and inhibitions?
3. Classification, detection, and segmentation
4. State of the art (SOTA) in object detection and segmentation
5. Data visualization
6. What is Detectron2?
7. What are the basic requirements for installing Detectron2?
8. Building and installation of Detectron2
9. Model building and training
10. Augmentation Techniques for Detectron2
11. Model inference
12. How to interpret the results obtained from Detectron2
13. Plotting the detection and segmentation results
14. Save the model